

HOMework 1

BASICS, CATEGORIES, GROUPS AND EXAMPLES

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Disclaimer: Generative AI was used to help me find a decent way to typeset a permutation and to determine whether I should say an element is an equivalence class “over” or “under” a relation.

1. Let $\text{Mat}_2(\mathbb{R})$ be the set of 2×2 matrices, and let $M = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$. Let

$$\mathcal{B} = \{X \in \text{Mat}_2(\mathbb{R}) \mid MX = XM\}.$$

Find necessary and sufficient conditions for $\begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \mathcal{B}$. Challenge: Find an $\mathbb{R}$ -basis of $\mathcal{B}$.

Proof. In general, for $X = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \in \text{Mat}_2(\mathbb{R})$, we have

$$XM = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} 2p + r & 2q + s \\ 2r & 2s \end{pmatrix}$$

and

$$MX = \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2p & p + 2q \\ 2r & r + 2s \end{pmatrix},$$

so the matrices X which satisfy $MX = XM$ (i.e. $X \in \mathcal{B}$) are exactly those which solve the following system of equations:

$$\begin{aligned} 2p + r &= 2p \\ 2q + s &= p + 2q \\ 2r &= 2r \\ 2s &= r + 2s. \end{aligned}$$

The third equation tells us nothing, but the first and fourth equations imply that $r = 0$, while the second says that $s = p$. This leaves us with $X = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix}$. This is necessary since

we derived it from the general form of a 2×2 matrix X , but it is also sufficient since plugging the values back into the system of equations yields

$$\begin{aligned} 2p + 0 &= 2p \\ 2q + p &= p + 2q \\ 0 &= 0 \\ 2p &= 0 + 2p. \end{aligned}$$

(Challenge) We will show that the matrices $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ form a basis for $\mathcal{B}$. For a given $C = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix} \in \mathcal{B}$, $C = pA + qB$, so the span of A and B contains $\mathcal{B}$. But any linear combination of A and B is in $\mathcal{B}$, so in fact their span is exactly $\mathcal{B}$. Finally, they are linearly independent since any scalar multiple of A has an upper-right entry of 0, so it cannot be B . Therefore, they form a basis for $\mathcal{B}$. $\square$

2. Determine if the following relations give well-defined functions:

(a) $f : \mathbb{Q} \rightarrow \mathbb{Z}$ given by $f(\frac{a}{b}) = a$.

(b) $f : \mathbb{Q} \rightarrow \mathbb{Q}$ given by $f(\frac{a}{b}) = \frac{a^2}{b^2}$.

Proof.

(a) Consider $a = 4, b = 6$. Then $f(\frac{a}{b}) = f(\frac{4}{6}) = 4$, but using the fact that $\frac{a}{b} = \frac{4}{6} = \frac{2}{3}$, we have $f(\frac{a}{b}) = f(\frac{2}{3}) = 2$. Thus f is not well-defined.

(b) Since $\frac{a^2}{b^2} = (\frac{a}{b})^2$, this is equivalent to saying that $f(x) = x^2$ for any $x \in \mathbb{Q}$. This is unambiguous, and in particular it is not dependent on the representation of x as a ratio of two integers like part (a), so it is well-defined. $\square$

3. Let X and Y be sets and let $f : X \rightarrow Y$ be a function. Show that the relation $a \sim_f b \iff f(a) = f(b)$ for $a, b \in X$ forms an equivalence relation on X , and that the equivalence classes are the nonempty fibers of f . Conversely, show that if X is a set and $\sim$ is an equivalence relation on X , construct a surjective function $f : X \rightarrow Y$ (for some Y) such that $\sim = \sim_f$.

Proof. If $a, b, c \in X$, then $f(a) = f(a)$, so $a \sim_f a$. Also, if $a \sim_f b$, then $f(a) = f(b)$ and $f(b) = f(a)$, so $b \sim_f a$. Finally, if $a \sim_f b$ and $b \sim_f c$, then $f(a) = f(b) = f(c)$, so $a \sim_f c$. In short, $\sim_f$ “inherits” its reflexivity, symmetry, and transitivity from “=”.

If $f^{-1}(\{y\})$ is nonempty for some $y \in Y$, then for $a, b \in f^{-1}(\{y\})$, we have $f(a) = f(b) = y$, so $a \sim_f b$. Conversely, given some $a, b \in X$ where $a \sim_f b$, take $y = f(a) = f(b)$. Then $a, b \in f^{-1}(\{y\})$, so the equivalence classes under $\sim_f$ are exactly the nonempty fibers of y .

Finally, given an equivalence relation $\sim$ on a set X , define $f : X \rightarrow X/\sim$ by $f(x) = [x]_\sim$ for any $x \in X$, where $[x]_\sim$ is the equivalence class of x under $\sim$. Then f is surjective since $[y]_\sim = f(y)$ for any $[y]_\sim \in X/\sim$.

If $a, b \in X$ such that $a \sim b$, then $f(a) = [a]_{\sim} = [b]_{\sim} = f(b)$, so $a \sim_f b$. On the other hand, if $a \sim_f b$, then $f(a) = f(b) = [b]_{\sim} = [a]_{\sim}$, so $a \sim b$ since they have the same equivalence class under $\sim$. $\square$

4. Find the greatest common divisor d of 69 and 372, and find a and b such that $69a + 372b = d$.

Proof. We use the Euclidean algorithm:

$$372 = 69 \cdot 5 + 27$$

$$69 = 27 \cdot 2 + 15$$

$$27 = 15 \cdot 1 + 12$$

$$15 = 12 \cdot 1 + 3$$

$$12 = 3 \cdot 4 + 0.$$

This tells us that the gcd is 3, and we can follow it back through the algorithm to get

$$\begin{aligned} 3 &= 15 - 12 \\ &= 15 - (27 - 15) \\ &= (69 - 27 \cdot 2) - (27 - (69 - 27 \cdot 2)) \\ &= (69 - (372 - 69 \cdot 5) \cdot 2) - ((372 - 69 \cdot 5) - (69 - (372 - 69 \cdot 5) \cdot 2)) \\ &= (69 - (372 - 69 \cdot 5) \cdot 2) - (372 - 69 \cdot 5 - 69 + (372 - 69 \cdot 5) \cdot 2) \\ &= (69 - 372 \cdot 2 + 69 \cdot 10) - (372 \cdot 3 - 69 \cdot 16) \\ &= 69 \cdot 11 - 372 \cdot 2 - 372 \cdot 3 + 69 \cdot 16 \\ &= 69 \cdot 27 - 372 \cdot 5. \end{aligned}$$

Thus $a = 27$ and $b = -5$. $\square$

5. Prove that if d divides n then $\phi(d)$ divides $\phi(n)$.

Proof. If d divides n , then $n = dm$ for some integer m . Write the prime factorization of n as

$$n = \prod_{i \in I} p_i^{a_i}$$

for some indexing set I . Because $d|n$, we can partition I into subsets J and K yielding sets of integers $\{b_j\}_{j \in J}$ and $\{c_k\}_{k \in K}$ such that $b_j = a_j$, $c_k < a_k$, and

$$d = \prod_{j \in J} p_j^{b_j} \prod_{k \in K} p_k^{c_k}.$$

Then we have

$$\begin{aligned}
\phi(n) &= \phi\left(\prod_{i \in I} p_i^{a_i}\right) \\
&= \prod_{i \in I} \phi(p_i^{a_i}) \\
&= \prod_{j \in J} \phi(p_j^{a_j}) \prod_{k \in K} \phi(p_k^{a_k}) \\
&= \prod_{j \in J} \phi(p_j^{b_j}) \prod_{k \in K} p_k^{a_k-1}(p_k-1) \\
&= \prod_{j \in J} \phi(p_j^{b_j}) \prod_{k \in K} p_k^{c_k-1} p_k^{a_k-c_k}(p_k-1) \\
&= \prod_{j \in J} \phi(p_j^{b_j}) \prod_{k \in K} \phi(p_k^{c_k}) \prod_{k \in K} p_k^{a_k-c_k} \\
&= \phi\left(\prod_{j \in J} p_j^{b_j} \prod_{k \in K} p_k^{c_k}\right) \prod_{k \in K} p_k^{a_k-c_k} \\
&= \phi(d) \prod_{k \in K} p_k^{a_k-c_k}.
\end{aligned}$$

Thus $\phi(d)$ divides $\phi(n)$. □

6. Prove that the equation $a^2 + b^2 = 3c^2$ has no solutions in nonzero integers a, b, c . Hint: First consider the equation mod 4 and think about what elements are squares mod 4. Iterate the conclusion you draw from this analysis to reach a contradiction.

Proof. Given $x \in \mathbb{Z}$, if x is even, then $x = 2n$ for some integer n , and $x^2 = 4n^2$, so $x^2 \equiv 0 \pmod{4}$. If x is odd, then $x = m + 1$ for some even m . Note that $2m \equiv 0 \pmod{4}$ since m is even, so

$$\begin{aligned}
x^2 &\equiv m^2 + 2m + 1 \pmod{4} \\
&\equiv 0 + 0 + 1 \pmod{4} \\
&\equiv 1 \pmod{4}.
\end{aligned}$$

In general, then, x^2 is congruent to either 0 or 1 mod 4 for any integer x .

Now consider $a^2 + b^2 = 3c^2 \pmod{4}$. For nonzero $a, b, c \in \mathbb{Z}$, since c^2 is congruent to either 0 or 1 mod 4, $3c^2$ is congruent to either 0 or 3 mod 4. The left-hand side of the equation mod 4 is either 0 + 0, 0 + 1, 1 + 0, or 1 + 1, so it cannot be 3 mod 4, and thus it must be 0. The only way for this to happen is to have $a^2 \equiv b^2 \equiv 0 \pmod{4}$, so we can factor out a 4 to get

$$4a_0^2 + 4b_0^2 = 3(4c_0^2)$$

where a_0, b_0 , and c_0 are integers. Dividing by 4 yields $a_0^2 + b_0^2 = 3c_0^2$, revealing another solution to the equation which, as we have shown, must be congruent to 0 mod 4 on both sides. Since the prime factorization of an integer is of finite length, however, we can repeat this process of dividing by 4 until we reach an equation where the sides are not congruent to 0 mod 4, which we have shown to be impossible. □

7. Let X be a set, and let $\mathcal{P}(X)$ be the power set of X (i.e. the set of all subsets of X). Let $\mathbf{C}$ be the category whose objects are the elements of $\mathcal{P}(X)$ and the morphisms in $\mathbf{C}$ from U to V is either the inclusion map $U \rightarrow V$ if $U \subseteq V$ and the empty set if $U \not\subseteq V$. Prove that $\mathbf{C}$ is indeed a category. Challenge: Generalize the construction given here to any partially ordered set $(X, \geq)$.

Proof. Given $A, B, C \in \mathbf{C}$, $f \in \text{Hom}(A, B)$, and $g \in \text{Hom}(B, C)$, then by definition $A \subseteq B$ and $B \subseteq C$, so $A \subseteq C$ and $\text{Hom}(A, C)$ is nonempty. We define, as we must, $g \circ f = \varphi$ where φ is the single element of $\text{Hom}(A, C)$. Now let $h \in \text{Hom}(C, D)$ for some object D . Then both $(h \circ g) \circ f$ and $h \circ (g \circ f)$ are in $\text{Hom}(A, D)$, but since $\text{Hom}(A, D)$ has only one element, they must be equal, making composition associative as desired.

Each set A is a subset of itself, so indeed $\text{Hom}(A, A)$ is nonempty for each $A \in \mathbf{C}$, and we call its single element the identity id_A . This acts as an identity with respect to composition because for any $A, B \in \mathbf{C}$ and $f \in \text{Hom}(A, B)$, $\text{id}_B \circ f \in \text{Hom}(A, B)$ and $f \circ \text{id}_A \in \text{Hom}(A, B)$, so just as in the preceding paragraph, we conclude that the two are equal to f since they all are in $\text{Hom}(A, B)$, which has just one element.

Challenge: All that is needed to define an analogous category for a general partially ordered set $(X, \geq)$ is to define $\mathbf{C}$ such that $\text{Obj}(\mathbf{C}) = X$ and $\text{Hom}(A, B)$ has a single element if $B \geq A$ but is otherwise empty for all $A, B \in \mathbf{C}$. The proof above then applies quite readily by replacing expressions like “ $A \subseteq B$ ” with “ $B \geq A$ ”. This is no accident, since the subset relation defines a natural partial ordering on the power set of any set. $\square$

8. Let G be a group. Explain how to encode the group law of G into a category $\mathbf{C}$ with one object $*$.

Proof. Let $\mathbf{C}$ be a category with one object A where there exists a bijection $\varphi : \text{Hom}(A, A) \rightarrow G$ satisfying $\varphi(g \circ f) = \varphi(f)\varphi(g)$. This suggests a group isomorphism, and indeed, the operation of composition makes $\text{Hom}(A, A)$ into a group, which we now prove.

The axioms of associativity and existence of identity are guaranteed by the definition of a category. To show that inverses exist, first observe that

$$\varphi(\text{id}_A) = \varphi(\text{id}_A \circ \text{id}_A) = \varphi(\text{id}_A)\varphi(\text{id}_A).$$

By cancellation, we see that $\varphi(\text{id}_A) = e$. Then let $f \in \text{Hom}(A, A)$, $a = \varphi(f)$, and $g = \varphi^{-1}(a^{-1})$. Then

$$\begin{aligned} \varphi(f \circ g) &= \varphi(g)\varphi(f) \\ &= \varphi(\varphi^{-1}(a^{-1}))\varphi(\varphi^{-1}(a)) \\ &= a^{-1}a \\ &= e. \end{aligned}$$

A similar proof shows that $\varphi(g \circ f) = e$ as well, so since φ is injective, $f \circ g = g \circ f = \text{id}_A$, showing that g is a two-sided inverse for f .

Thus $\text{Hom}(A, A)$ is a group under composition, and φ is an isomorphism between $\text{Hom}(A, A)$ and G , showing that G can be encoded as a category with one object. $\square$

9. Prove that if $x^2 = 1$ for all $x \in G$, then G is abelian.

Proof. Indeed, $x = x^{-1}$ for all $x \in G$ since $x * x = 1$, so given $g, h \in G$,

$$\begin{aligned} gh &= (gh)^{-1} \\ &= h^{-1}g^{-1} \\ &= hg. \end{aligned}$$

□

10. Consider the group G with presentation

$$G = \langle u, v \mid u^4 = v^3 = 1, uv = v^2u^2 \rangle.$$

- (a) Show that $v^2 = v^{-1}$.
- (b) Show that v commutes with u^3 .
- (c) Show that v commutes with u .
- (d) Show that $uv = 1$.
- (e) Show that $u = 1$, deduce that $v = 1$, so that this is just a long-winded description of the trivial group!

Proof. (a)

$$\begin{aligned} v^3 = 1 &\implies v^3 * v^{-1} = 1 * v^{-1} \\ &\implies v^2 * (v * v^{-1}) = v^{-1} \\ &\implies v^2 * 1 = v^{-1} \\ &\implies v^2 = v^{-1} \end{aligned}$$

(b) First, observe that

$$\begin{aligned} v^2u^3v &= (v^2u^2)uv \\ &= (uv)uv \\ &= uv(uv) \\ &= uv(v^2u^2) \\ &= uv^3u^2 \\ &= u^3. \end{aligned}$$

Multiplying by v on the left, we get

$$\begin{aligned} v^3u^3v &= vu^3 \\ \implies u^3v &= vu^3. \end{aligned}$$

(c) First, we establish that

$$\begin{aligned} u^9 &= (u^8)u \\ &= (u^4 u^4)u \\ &= u, \end{aligned}$$

from which we derive

$$\begin{aligned} vu &= vu^9 \\ &= (vu^3)u^3u^3 \\ &= u^3(vu^3)u^3 \\ &= u^3u^3(vu^3) \\ &= u^3u^3u^3v \\ &= u^9v \\ &= uv. \end{aligned}$$

(d) Since u and v commute, we can transform the “commuting” relation like so:

$$\begin{aligned} uv &= v^2u^2 \\ &= u^2v^2 \\ &= u(uv)v \\ &= (uv)(uv) \\ &= (uv)^2. \end{aligned}$$

Multiplying by $(uv)^{-1}$ on both sides yields $1 = uv$.

(e) Indeed,

$$\begin{aligned} u &= u(1)^3 \\ &= u(uv)^3 \\ &= u^4v^3 \\ &= 1 \cdot 1 \\ &= 1. \end{aligned}$$

Then since $uv = 1$, substituting 1 for u yields $v = 1$.

Rewriting the presentation in light of this, we have

$$\begin{aligned} G &= \langle u, v \mid u^4 = v^3 = 1, uv = v^2u^2 \rangle \\ &= \langle 1, 1 \mid 1^4 = 1^3 = 1, 1 \cdot 1 = 1^21^2 \rangle \\ &= \langle 1 \rangle \end{aligned}$$

where the third equality comes from removing duplicate generators and trivial relations. Thus, G is the trivial group. □

11. Suppose that a and b are commuting elements of a group G such that $a^k = b^l$ implies that k is a multiple of the order of a , and l is a multiple of the order of b . Show that the order of ab is the least common multiple of $|a|$ and $|b|$. Be sure to use the first problem to help simplify $(ab)^n$. Also, it may help to review basic properties of lcms from chapter 0 in the text.

Proof. We will show that $(ab)^q = e$ for all q where q is a multiple of both $|a|$ and $|b|$, and in fact all powers p such that $(ab)^p = e$ are common multiples of $|a|$ and $|b|$. Then, having shown that the powers of ab which yield the identity are exactly the common multiples of $|a|$ and $|b|$, we conclude that the least of them, and thus the order of ab , is $\text{lcm}(|a|, |b|)$.

First, let $q \in \mathbb{Z}$ be a common multiple of $|a|$ and $|b|$, and write $q = m|a| = n|b|$ for some $m, n \in \mathbb{Z}$. Then

$$\begin{aligned} (ab)^q &= a^q b^q \\ &= a^{m|a|} b^{n|b|} \\ &= (a^{|a|})^m (b^{|b|})^n \\ &= e^m e^n \\ &= e. \end{aligned}$$

Now assume $p \in \mathbb{Z}$ such that $(ab)^p = e$. Then

$$\begin{aligned} a^p b^p &= e \\ \implies a^p &= b^{-p}, \end{aligned}$$

and by hypothesis, this means that there exist $m, n \in \mathbb{Z}$ such that $p = m|a|$ and $-p = n|b|$. Thus p is a multiple of both $|a|$ and $|b|$ as desired. □

12. Let σ be the 12-cycle $(1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12)$. For all $i \geq 1$, write the cycle decomposition of σ^i . For which positive integers i is σ^i also a 12-cycle? Formulate a result about when a power of an n -cycle is also an n -cycle. (If you are having trouble coming up with what to prove here, try out cycles of different lengths). Challenge: Prove your claim.

Proof.

$$\begin{aligned}
\sigma &= (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12) \\
\sigma^2 &= (1\ 3\ 5\ 7\ 9\ 11)(2\ 4\ 6\ 8\ 10\ 12) \\
\sigma^3 &= (1\ 4\ 7\ 10)(2\ 5\ 8\ 11)(3\ 6\ 9\ 12) \\
\sigma^4 &= (1\ 5\ 9)(2\ 6\ 10)(3\ 7\ 11)(4\ 8\ 12) \\
\sigma^5 &= (1\ 6\ 11\ 4\ 9\ 2\ 7\ 12\ 5\ 10\ 3\ 8) \\
\sigma^6 &= (1\ 7)(2\ 8)(3\ 9)(4\ 10)(5\ 11)(6\ 12) \\
\sigma^7 &= (1\ 8\ 3\ 10\ 5\ 12\ 7\ 2\ 9\ 4\ 11\ 6) \\
\sigma^8 &= (1\ 9\ 5)(2\ 10\ 6)(3\ 11\ 7)(4\ 12\ 8) \\
\sigma^9 &= (1\ 10\ 7\ 4)(2\ 11\ 8\ 5)(3\ 12\ 9\ 6) \\
\sigma^{10} &= (1\ 11\ 9\ 7\ 5\ 3)(2\ 12\ 10\ 8\ 6\ 4) \\
\sigma^{11} &= (1\ 12\ 11\ 10\ 9\ 8\ 7\ 6\ 5\ 4\ 3\ 2) \\
\sigma^{12} &= (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12),
\end{aligned}$$

and in general $\sigma^i = \sigma^{i \bmod 12}$. Let $\bar{a}$ denote the equivalence class represented by a in $\mathbb{Z}/12\mathbb{Z}$. Then σ^i is a 12-cycle for $i = \bar{1}, \bar{5}, \bar{7}, \bar{11}$. I would conjecture, then, that a power of an n -cycle τ , say τ^j , is an n -cycle if and only if $\gcd(j, n) = 1$.

(Challenge) In general, the i -th power of an n -cycle σ sends an element appearing at index j in the cycle to the entry at index $(j + i \bmod n)$. Thus, σ^i is an n -cycle exactly when the process of adding $i \bmod n$ starting at any element “hits every other element” of σ . This happens whenever $\gcd(i, n) = 1$, and in general σ^i will have $\gcd(i, n)$ cycles. $\square$